

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
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        "required": false,
        "schema": {
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          "type": "string"
        },
        "chamber": {
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          "required": false,
          "schema": {
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          },
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            "required": false,
            "schema": {
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              "type": "string"
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            "required": false,
            "schema": {
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            "required": false,
            "schema": {
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              "type": "boolean"
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            "schema": {
              "type": "string"
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            "required": true,
            "schema": {
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              "type": "string"
            },
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                    "fields": {
                      "action_code": {
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                        "required": false,
                        "schema": {
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                          "type": "string"
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                      "alarm_type": {
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                        "required": true,
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                          "type": "string"
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                        "required": false,
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                        "required": true,
                        "schema": {
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                        "schema": {
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                        "required": true,
                        "schema": {
                          "type": "string"
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                        "nullable": true,
                        "required": false,
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                          "type": "string"
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                        "required": true,
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                        "required": true,
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                          "type": "string"
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                        "required": true,
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                        "schema": {
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                          "type": "string"
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                        "required": false,
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                          "type": "string"
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                          "type": "string"
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                        "schema": {
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                          "type": "integer"
                        },
                      "rule_code": {
                        "nullable": false,
                        "required": true,
                        "schema": {
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                        "schema": {
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                          "type": "integer"
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "integer"
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                        "required": false,
                        "schema": {
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                        },
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                        },
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                        "nullable": false,
                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
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                              "required": false,
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}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
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          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
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          "type": "string"
        }
      }
    },
    "responses": {
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        "schema": {
          "items": {
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            "fields": {
              "measured_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
              "value": {
                "nullable": false,
                "required": true,
                "schema": {
                  "type": "number"
                }
              },
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              "type": "array",
              "shape": "array",
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                      "required": false,
                      "schema": {
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                          "fields": {
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                              "required": true,
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                                    {
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                                  "type": "array",
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                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

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s": {"additional_properties": false, "fields": {"LCL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "LSL": {"nullable": true, "required": false, "schema": {"type": "number"}}, "TARGET": {"nullable": false, "required": true, "schema": {"type": "number"}}, "UCL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "USL": {"nullable": false, "required": true, "schema": {"type": "number"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "ctrl_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "ctrl_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "spec_lower": {"nullable": true, "required": false, "schema": {"type": "number"}}, "spec_upper": {"nullable": false, "required": true, "schema": {"type": "number"}}, "target_value": {"nullable": false, "required": true, "schema": {"type": "number"}}, "unit": {"nullable": true, "required": false, "schema": {"type": "string"}}, "upper_only": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object", "type": "array", "shape": "array"}}
```

04 POST /documents/search

RULES []

```
{ "request": { "body": {"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer"}}, "type": "object", "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": -1.0, "type": "number"}}, "section": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "array": true, "shape": "array", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}}
```

05 GET /relations/chambers/{chamber_id}

RULES [{"node_count equals len(nodes)", "relationship_count equals len(relationships)"]

```
{ "request": { "body": null, "header": {}, "path": {"chamber_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "type": "object", "type": "array", "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "type": "array", "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array", "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object", "type": "array", "type": "object", "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id":
```

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{
  "nullable": false, "required": true, "schema": {
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  }, "confidence": {
    "nullable": true, "required": true, "schema": {
      "maximum": 1.0, "minimum": 0.0, "type": "number"
    }, "created_at": {
      "nullable": false, "required": true, "schema": {
        "format": "date-time", "type": "string"
      }, "deliveries": {
        "nullable": false, "required": true, "schema": {
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                    "EMAIL", "MES"
                  ], "type": "string"
                }, "status": {
                  "nullable": false, "required": true, "schema": {
                    "enum": [
                      "BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"
                    ], "type": "string"
                  }, "type": "object", "type": "array"
                }, "fault_code": {
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                      "FOC", "MFD", "OTH", "RFM", "TMD"
                    ], "type": "string"
                  }, "fault_color": {
                    "nullable": false, "required": true, "schema": {
                      "type": "null"
                    }, "fault_name": {
                      "nullable": false, "required": true, "schema": {
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                      }, "latency_ms": {
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                          }, "predicted_fault_code": {
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                            }, "recommended_action": {
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                              }, "status": {
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                                          }, "result_summary": {
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                                              }, "shape": "array", "422": {
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                                                                    }, "type": "array"
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```

```
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```

08 GET /approvals

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields null", "only EQP_HOLD actions have approvals"]

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {} }, "responses": { "200": { "schema": { "items": { "additional_properties": false, "fields": { "action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "type": "array", "shape": "array" } }
```

09 POST /approvals/{approval_id}/decision

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields null", "only EQP_HOLD actions have approvals"]

```
{ "request": { "body": { "additional_properties": false, "fields": { "decided_by": {"nullable": false, "required": true, "schema": {"max_length": 40, "min_length": 1, "type": "string"}}, "decision": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "REJECTED"], "type": "string"}}, "decision_comment": {"nullable": true, "required": false, "schema": {"max_length": 1000, "type": "string"}}, "type": "object"}, "header": {}, "path": { "approval_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {} }, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable":
```

```
: true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable":
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ength": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["F
OC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schem
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{"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required
": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true,
"schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape":
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ype": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "a
rray"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
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ctor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": tr
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able": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "t
ype": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": fal
se, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "re
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, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "req
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```

```

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13 GET /alarms/{source}/{alarm_id}
RULES []
{"request": {"body": null, "header": {}, "path": {"alarm_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "agent_run_status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": false, "schema": {"enum": ["APPROVED", "AUTO", "EXPIRED", "PENDING", "REJECTED"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "detail": {"nullable": false, "required": true, "schema": {"type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "hit_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "judgement": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "latest agent

```

```
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```

14 GET /alarms/paged

RULES []

```
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15 GET /traces/catalog

RULES []

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16 POST /traces/search

RULES []

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17 GET /relations/equipment/{equipment_id}

RULES []

deferred inventory

18 GET /documents/{document_id}

RULES []

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```

22 POST /agent/runs/{run_id}/retry

RULES []

deferred inventory

23 GET /agent/runs/paged

RULES []

deferred inventory

24 GET /approvals/paged

RULES []

deferred inventory

25 GET /actions

RULES []

```
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```

```
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```

26 GET /actions/{action_id}

RULES []

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CY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UN
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```

27 POST /actions/{action_id}/deliveries/{channel}/retry

RULES []

deferred inventory

28 POST /internal/actions/{action_id}/delivery

RULES [{"FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provider_message_id non-null and error_code null"}]

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      "fields": {
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          "required": true,
          "schema": {
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            "type": "string"
          }
        },
        "completed_at": {
          "nullable": false,
          "required": true,
          "schema": {
            "format": "date-time",
            "type": "string"
          }
        },
        "error_code": {
          "nullable": true,
          "required": true,
          "schema": {
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            "type": "string"
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        },
        "event_id": {
          "nullable": false,
          "required": true,
          "schema": {
            "min_length": 1,
            "type": "string"
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        },
        "provider_message_id": {
          "nullable": true,
          "required": true,
          "schema": {
            "min_length": 1,
            "type": "string"
          }
        },
        "request_hash": {
          "nullable": false,
          "required": true,
          "schema": {
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            "min_length": 64,
            "pattern": "^([0-9a-f]{64})$",
            "type": "string"
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        },
        "status": {
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          "required": true,
          "schema": {
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            "type": "string"
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        "type": "object",
        "header": {
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            "required": false,
            "schema": {
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            "required": false,
            "schema": {
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                    },
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                      "required": true,
                      "schema": {
                        "enum": ["EMAIL", "MES MOCK"],
                        "type": "string"
                      }
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                    "completed_at": {
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                }
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    }
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}
```

29 POST /analytics/query

RULES []

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{
  "request": {
    "body": {
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      "fields": {
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          "nullable": false,
          "required": true,
          "schema": {
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            "min_length": 1,
            "type": "string"
          }
        },
        "type": "object",
        "header": {},
        "path": {},
        "query": {},
        "responses": {
          "200": {
            "schema": {
              "additional_properties": false,
              "fields": {
                "columns": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "items": {
                      "type": "string"
                    },
                    "type": "array"
                  },
                  "cross_check": {
                    "nullable": true,
                    "required": false,
                    "schema": {
                      "additional_properties": false,
                      "fields": {
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                          "nullable": true,
                          "required": false,
                          "schema": {
                            "type": "string"
                          }
                        },
                        "status": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "enum": ["MATCH", "MISMATCH", "SKIPPED"],
                            "type": "string"
                          }
                        }
                      },
                      "summary": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "string"
                        }
                      },
                      "type": "object",
                      "error_msg": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "string"
                        }
                      },
                      "generated_sql": {
                        "nullable": true,
                        "required": false,
                        "schema": {
                          "type": "string"
                        }
                      },
                      "group_by": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "items": {
                            "type": "string"
                          },
                          "type": "array"
                        },
                        "is_rejected": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "type": "boolean"
                          }
                        },
                        "is_valid": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "type": "boolean"
                          }
                        },
                        "latency_ms": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "minimum": 0.0,
                            "type": "integer"
                          }
                        },
                        "metric": {
                          "nullable": true,
                          "required": false,
                          "schema": {
                            "additional_properties": false,
                            "fields": {
                              "column": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "type": "string"
                                }
                              },
                              "p": {
                                "nullable": true,
                                "required": false,
                                "schema": {
                                  "maximum": 100.0,
                                  "minimum": 0.0,
                                  "type": "number"
                                }
                              },
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "enum": ["count", "max", "mean", "median", "min", "percentile", "ratio", "std", "sum"],
                                "type": "string"
                              },
                              "type": "object",
                              "metric_result": {
                                "nullable": false,
                                "required": false,
                                "schema": {
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                                  "variants": [
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                                      "type": "integer"
                                    },
                                    {
                                      "type": "number"
                                    }
                                  ],
                                  "items": {
                                    "additional_properties": false,
                                    "fields": {
                                      "group": {
                                        "nullable": false,
                                        "required": true,
                                        "schema": {
                                          "additional_properties": true,
                                          "fields": {},
                                          "type": "object"
                                        },
                                          "value": {
                                            "nullable": false,
                                            "required": false,
                                            "schema": {
                                              "type": "union",
                                              "variants": [
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                                                  "type": "integer"
                                                },
                                                {
                                                  "type": "number"
                                                }
                                              ],
                                              "type": "null"
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                                          }
                                        },
                                        "type": "array",
                                        "type": "null"
                                      },
                                      "nl_query_log_id": {
                                        "nullable": true,
                                        "required": false,
                                        "schema": {
                                          "minimum": 1.0,
                                          "type": "integer"
                                        }
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                                      "question": {
                                        "nullable": false,
                                        "required": true,
                                        "schema": {
                                          "type": "string"
                                        }
                                      },
                                      "reject_reason": {
                                        "nullable": true,
                                        "required": false,
                                        "schema": {
                                          "type": "string"
                                        }
                                      },
                                      "row_count": {
                                        "nullable": false,
                                        "required": true,
                                        "schema": {
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                                          "type": "integer"
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                                        "required": true,
                                        "schema": {
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                                            "additional_properties": true,
                                            "fields": {},
                                            "type": "object"
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                                          "type": "array"
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                                          "visualization": {
                                            "nullable": true,
                                            "required": false,
                                            "schema": {
                                              "additional_properties": false,
                                              "fields": {
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                                                  "nullable": false,
                                                  "required": true,
                                                  "schema": {
                                                    "enum": ["bar", "histogram", "line", "table"],
                                                    "type": "string"
                                                  }
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                                                  "required": false,
                                                  "schema": {
                                                    "type": "string"
                                                  }
                                                },
                                                "y": {
                                                  "nullable": true,
                                                  "required": false,
                                                  "schema": {
                                                    "type": "string"
                                                  }
                                                }
                                              },
                                              "type": "object",
                                              "shape": "object",
                                              "422": {
                                                "schema": {
                                                  "additional_properties": true,
                                                  "fields": {
                                                    "detail": {
                                                      "nullable": false,
                                                      "required": false,
                                                      "schema": {
                                                        "items": {
                                                          "additional_properties": true,
                                                          "fields": {
                                                            "loc": {
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                                                              "required": true,
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                                                                  "variants": [
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                                                                      "type": "string"
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                                                                    {
                                                                      "type": "integer"
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                                                                  "type": "array"
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                                                                "msg": {
                                                                  "nullable": false,
                                                                  "required": true,
                                                                  "schema": {
                                                                    "type": "string"
                                                                  }
                                                                },
                                                                "type": {
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                                                                  "required": true,
                                                                  "schema": {
                                                                    "type": "string"
                                                                  }
                                                                }
                                                              },
                                                              "type": "object",
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                                                            },
                                                            "shape": "object"
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                                                        }
                                                    }
                                                  }
                                                }
                                              }
                                            }
                                          }
                                        }
                                      }
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

30 POST /analytics/graph-query

RULES []

```
{
  "request": {
    "body": {
      "additional_properties": false,
      "fields": {
        "question": {
          "nullable": false,
          "required": true,
          "schema": {
            "max_length": 1000,
            "min_length": 1,
            "type": "string"
          }
        },
        "type": "object",
        "header": {}
      }
    }
  }
}
```

```
": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
```

31 POST /analytics/validate

RULES []

```
{ "request": { "body": { "additional_properties": false, "fields": { "sql": { "nullable": false, "required": true, "schema": { "max_length": 20000, "min_length": 1, "type": "string"}}, "type": "object"}, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"checks": {"nullable": true, "required": false, "schema": {"items": {"additional_properties": false, "fields": {"key": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "ok": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "type": "array"}}, "normalized_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"type": "string"}}, "valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
```

32 GET /analytics/history

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "is_rejected": {"nullable": true, "required": false, "schema": {"type": "boolean"}}, "is_valid": {"nullable": true, "required": false, "schema": {"type": "boolean"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}
```

33 GET /analytics/evaluations

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "latest": {"nullable": false, "required": false, "schema": {"default": true, "type": "boolean"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"accuracy": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "correct": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "defense_passed": {"nullable": false, "required":
```

```
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```

34 GET /audit-logs/paged

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "page": {"nullable": false, "required": false, "schema": {"default": 1, "minimum": 1, "type": "integer"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}, "responses": {"200": {"schema": {"additional_properties": true, "fields": {"event_type_counts": {"nullable": false, "required": true, "schema": {"additional_properties": {"type": "integer"}, "fields": {}, "type": "object"}}, "event_types": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, {"type": "array"}, {"page": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"type": "integer"}}, {"type": "object"}, {"shape": "object"}, {"422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": "object"}, {"type": "array"}}, {"type": "object"}, {"shape": "object"}}}
```

35 GET /health

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"status": {"nullable": false, "required": true, "schema": {"type":
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[illegible]

```
EADY", "READY"], "type": "string"}}, "type": "object"), "shape": "object"}}
DTO COMPONENTS
ActionCode
{"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}
ActionDeliveryDetailItem
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "completed_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "started_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}
ActionDeliveryItem
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}
ActionDetailResponse
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AgentRunItem

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AlarmAskEvidence

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AlarmEvidence

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AlarmItem

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      }
    },
    "alarm_id": {
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      "required": true,
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        "type": "string"
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    "alarm_type": {
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      "required": true,
      "schema": {
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        "type": "string"
      }
    },
    "area": {
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        "type": "string"
      }
    },
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      }
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    },
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        "type": "string"
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        "type": "string"
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    },
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    },
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    },
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    "recipe_step_no": {
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        "type": "integer"
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    "rule_code": {
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      "required": true,
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        "type": "string"
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      "schema": {
        "type": "number"
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    },
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    "wafer_id": {
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      "required": true,
      "schema": {
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        "type": "string"
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    },
    "rules": {
      "source TRACE requires rule_code TRACE_OOS, source SUMMARY requires rule_code SUMMARY_OOC, source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null",
      "compatibility aliases derive only from canonical fields",
      "type": "object"
    }
  }
}
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AlarmRef

```
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  "additional_properties": false,
  "fields": {
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      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "source": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["R03", "SUMMARY", "TRACE"],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

AlarmSource

```
{
  "enum": ["R03", "SUMMARY", "TRACE"],
  "type": "string"
}
```

AnalysisQueryRequest

```
{
  "additional_properties": false,
  "fields": {
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      "required": true,
      "schema": {
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        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

AnalysisQueryResponse

```
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      "schema": {
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    },
    "cross_check": {
      "nullable": true,
      "required": false,
      "schema": {
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        "fields": {
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            "nullable": true,
            "required": false,
            "schema": {
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            }
          },
          "status": {
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            "required": true,
            "schema": {
              "enum": ["MATCH", "MISMATCH", "SKIPPED"],
              "type": "string"
            }
          },
          "summary": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
            }
          },
          "error_msg": {
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            }
          },
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            }
          },
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              "type": "array"
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                    }
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                "type": "object"
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              "type": "string"
            }
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            }
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              },
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                "schema": {
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                "schema": {
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                }
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                    }
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                      "type": "string"
                    }
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            }
          }
        }
      }
    }
  }
}
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FaultHypothesis

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  "type": "string"
}
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GraphAskEvidence

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    "graph_revision": {
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        "type": "string"
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  "type": "object"
}
```

GraphContext

```
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      "schema": {
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        },
        "type": "array"
      }
    },
    "area": {
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      "required": true,
      "schema": {
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        "type": "string"
      }
    },
    "chamber_id": {
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      }
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    "equipment_id": {
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        "type": "string"
      }
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        },
        "type": "array"
      }
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    "process_step_id": {
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        "type": "string"
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    },
    "relation_ids": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
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        },
        "type": "array"
      }
    }
  },
  "type": "object"
}
```

GraphEvidence

```
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      }
    },
    "graph_revision": {
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      "required": true,
      "schema": {
        "pattern": "^[0-9a-f]{64}$",
        "type": "string"
      }
    },
    "relation_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "pattern": "^REL-[0-9a-f]{20}$",
        "type": "string"
      }
    },
    "source_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "title": {
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      "required": true,
      "schema": {
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    },
    "type": {
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      "required": true,
      "schema": {
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        "type": "string"
      }
    }
  },
  "type": "object"
}
```

GraphNode

```
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  "fields": {
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      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "label": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"],
        "type": "string"
      }
    },
    "name": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "node_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "properties": {
      "nullable": false,
      "required": true,
      "schema": {
        "additional_properties": true,
        "fields": {},
        "type": "object"
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    }
  },
  "type": "object"
}
```

GraphQueryRequest

```
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  "additional_properties": false,
  "fields": {
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      "required": true,
      "schema": {
        "max_length": 1000,
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

GraphQueryResponse

```
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  "additional_properties": false,
  "fields": {
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      "required": true,
      "schema": {
        "items": {
          "type": "string"
        },
        "type": "array"
      }
    },
    "error_msg": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "generated_cypher": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "is_rejected": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "boolean"
      }
    },
    "is_valid": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "boolean"
      }
    },
    "latency_ms": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 0.0,
        "type": "integer"
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    },
    "question": {
      "nullable": false,
      "required": true,
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    }
  },
  "type": "object"
}
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HealthResponse
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ImpactScopeBlock
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NIQueryOutcome
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  "type": "object"
}

SqlValidateResponse
{
  "additional_properties": false,
  "fields": {
    "checks": {
      "nullable": true,
      "required": false,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "key": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "label": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              }
            },
            "ok": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              }
            }
          },
          "type": "object"
        },
        "type": "array"
      },
      "normalized_sql": {
        "nullable": true,
        "required": false,
        "schema": {
          "type": "string"
        }
      },
      "reason": {
        "nullable": false,
        "required": true,
        "schema": {
          "type": "string"
        }
      },
      "valid": {
        "nullable": false,
        "required": true,
        "schema": {
          "type": "boolean"
        }
      }
    }
  },
  "type": "object"
}

ToolCallStatus
{
  "enum": [
    "ERROR",
    "SUCCESS",
    "TIMEOUT"
  ],
  "type": "string"
}

TraceAskEvidence
{
  "additional_properties": false,
  "fields": {
    "excerpt": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "source_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "title": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    }
  },
  "type": "object"
}

TraceEvidence
{
  "additional_properties": false,
  "fields": {
    "excerpt": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "source_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "title": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "TRACE"
        ],
        "type": "string"
      }
    }
  },
  "type": "object"
}

TracePoint
{
  "additional_properties": false,
  "fields": {
    "measured_at": {
      "nullable": false,
      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      }
    },
    "recipe_step_no": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 1,
        "type": "integer"
      }
    },
    "seq_no": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 0,
        "type": "integer"
      }
    },
    "value": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    }
  },
  "type": "object"
}

ValidationCheck
{
  "additional_properties": false,
  "fields": {
    "key": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "label": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "ok": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "boolean"
      }
    }
  },
  "type": "object"
}

ValidationError
{
  "additional_properties": true,
  "fields": {
    "loc": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "type": "union",
          "variants": [
            {
              "type": "string"
            },
            {
              "type": "integer"
            }
          ],
          "type": "array"
        },
        "msg": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "string"
          }
        },
        "type": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "string"
          }
        }
      }
    }
  },
  "type": "object"
}

VisualizationPlan
{
  "additional_properties": false,
  "fields": {
    "chart_type": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "bar",
          "histogram",
          "line",
          "table"
        ],
        "type": "string"
      }
    },
    "x": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "y": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    }
  },
  "type": "object"
}

AUDIT EVENTS
DETECTION_COMPLETED / LOT_HIST / A
AGENT_RUN_STARTED / AGENT_RUN / C
HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
AGENT_RUN_COMPLETED / AGENT_RUN / C
AGENT_RUN_FAILED / AGENT_RUN / C
```